A VQE–QSE Workflow for Ground- and Excited-State Simulation of Ethylene

1 Introduction

Quantum computers represent and manipulate quantum states directly, offering an alternative to classical approximations for problems governed by quantum mechanics — making quantum chemistry, where electronic structure is inherently quantum-mechanical, a natural application. Excited-state calculations are especially demanding: photo-induced transitions of electrons between orbitals underlie photochemistry, photosynthesis, and photovoltaics, but the number of possible electronic configurations grows rapidly with system size, straining classical methods.

This work uses ethylene (C_2H_4) — chosen for its well-characterized $\pi \to \pi^*$ excitation and small electronic structure — as a testbed, examining both its equilibrium and 90° twisted geometries to probe how structure affects electronic states. The workflow combines the Variational Quantum Eigensolver (VQE), which optimizes a quantum circuit to approximate the ground-state energy, with Quantum Subspace Expansion (QSE), which builds on that ground state to extract low-lying excited-state energies. The goal is to build and evaluate this VQE–QSE pipeline for ethylene, benchmarking results against classical methods and literature values while assessing the impact of ansatz choice and quantum noise — and to position the workflow as a foundation for larger photoactive systems, where excitation energies could feed electron-transfer models relevant to photosynthesis and solar-energy conversion.

2 Computational Methodology

2.1 Molecular System and Active Space

Ethylene (C_2H_4) was studied at its equilibrium and 90° twisted geometries. The STO-3G basis set was used with an active space of 2 electrons in 2 spatial orbitals. This reduced the molecular problem to a manageable size for quantum simulation.

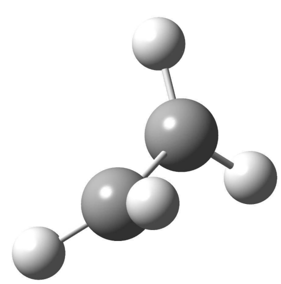


Figure 1: Molecular structure of ethylene

2.2 VQE Ground-State Calculation

The Variational Quantum Eigensolver (VQE) was used to estimate the ground-state energy. Both UCCSD and hardware-efficient ansätze were tested. The Jordan-Wigner mapping required 4 qubits for the selected active space, while parity mapping with tapering reduced the problem to 2 qubits.

2.3 QSE Excited-State Calculation

The optimized VQE ground state was used as the starting point for Quantum Subspace Expansion (QSE). Excitation operators were applied to generate a small subspace of possible excited states:

$$|\psi_i\rangle = O_i|\psi_0\rangle$$

The Hamiltonian and overlap matrices were constructed in this subspace, and the generalized eigenvalue problem

$$HC = SCE$$

where H is the Hamiltonian matrix in the subspace, S is the overlap matrix of the (non-orthogonal) basis, C contains the eigenvector coefficients, and E is the diagonal matrix of excited-state energies (eigenvalues). was solved to obtain the low-lying excited-state energies.

2.4 Noise and Classical Benchmarking

The effect of quantum noise was studied using a noisy simulator with depolarizing and readout errors. Noise mitigation was investigated using ZNE-style extrapolation and readout mitigation.

For comparison, classical CASCI, CASSCF, and EOM-CCSD calculations were performed for the same molecular system.

3 Results and Discussion

3.1 Ground-State VQE Results

The VQE results were close to the exact active-space ground-state energy of (-77.116593) Ha as shown in figure 2. UCCSD reached the exact value, while EfficientSU2 and RealAmplitudes showed only small deviations.

3.2 QSE Excited-State Results

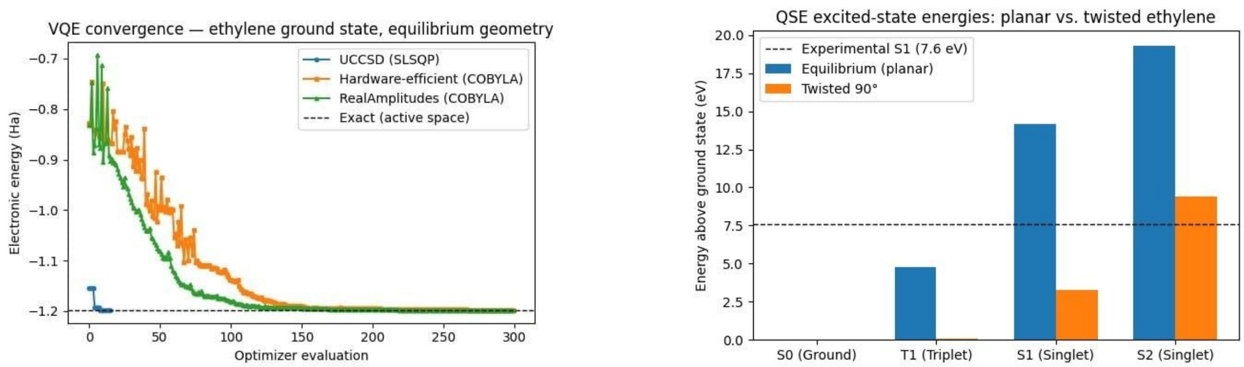
QSE was used to calculate the low-lying excited states from the optimized VQE ground state. At the equilibrium geometry with STO-3G, the excitation energies were 4.792, 14.155, and 19.307 eV. The calculated $\pi \to \pi^*$ excitation was 14.155 eV, compared with the reference value of 7.6 eV.

3.3 Effect of Geometry and Basis Set

The excitation energies were strongly affected by both molecular geometry and basis set. The $\pi \to \pi^*$ excitation decreased from 14.155 eV with STO-3G to 7.565 eV with aug-cc-pVDZ, which is close to the 7.6 eV reference. At the 90° twisted geometry, the low-lying excitation energies were also significantly reduced, with values of 0.084, 3.247, and 9.401 eV (STO-3G). With the aug-cc-pVDZ basis set, the twisted-geometry excitation energies were 0.053, 2.862, and 3.424 eV.

3.4 Noise and Error Mitigation

A noisy simulation was performed using depolarizing and readout errors. At $1\times$ noise, the energy error was 177.20 mHa. ZNE-style extrapolation reduced the error to 94.04 mHa, recovering about 47% of the noise-induced error. Readout mitigation was also implemented.



(a) VQE convergence

(b) QSE excitation energies

Figure 2: VQE and QSE results for ethylene

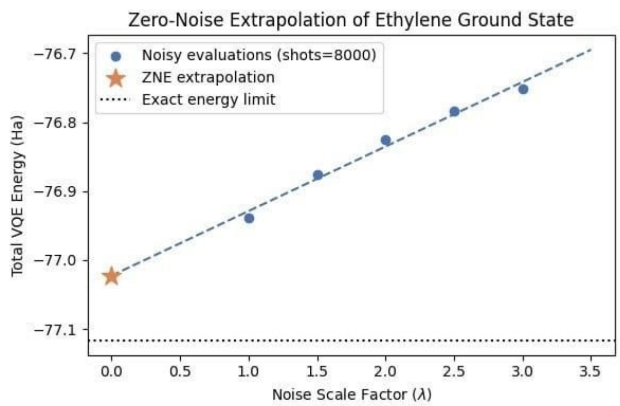


Figure 3: Zero-noise extrapolation of the ethylene ground-state energy toward the exact limit

3.5 Classical Benchmarking

The quantum results were compared with classical methods. CASCI and CASSCF gave similar ground-state and excitation energies, while EOM-CCSD produced a lower excitation energy due to its different treatment of electron correlation.

Method	Ground Energy (Ha)	$S_0 \to S_1$ (eV)
CASCI	-77.116593	14.155
CASSCF	-77.116475	14.142
EOM-CCSD	-77.234131	11.609

Table 1: Comparison of ground-state energies and excitation energies obtained from classical methods.

4 Extension Toward Photo-Induced Charge Transfer

4.1 Donor–Acceptor Model

As a simple extension toward photosynthetic charge transfer, two ethylene molecules were modeled as a donor–acceptor pair. A 4-electron/4-orbital active space was used to describe the interacting system, providing a minimal model for studying excitation and charge-transfer processes.

4.2 Electronic Coupling

The electronic coupling between the donor and acceptor was estimated from the energy splitting of the lowest excited states. This coupling was then used as an input for modeling the electron-transfer process.

4.3 Marcus Electron Transfer

Marcus theory was used to estimate the electron-transfer rate from the calculated electronic coupling. A reorganization energy of 0.25 eV was used, and the rate was evaluated over a range of driving forces at 298 K.

4.4 Lindblad Dynamics

Lindblad dynamics was used to model the population transfer between donor, acceptor, charge-transfer, and ground states. The model includes charge-transfer loss and dephasing to describe the dynamics of the photo-induced electron-transfer process.

4.5 Future Direction: Quantum Photo-Conversion Score

A possible future development of this work is to combine the electronic-structure and charge-transfer results into a single screening metric for photoactive molecular systems. We propose a preliminary Quantum Photo-Conversion Score (QPCS) that combines the suitability of the excitation with the efficiency of electron transfer:

$$\mathrm{QPCS} = \eta_{\mathrm{exc}} \frac{k_{\mathrm{ET}}}{k_{\mathrm{ET}} + k_{\mathrm{loss}}} \tag{1}$$

where η_{exc} represents the suitability of the calculated excitation energy for the target application, k_{ET} is the Marcus electron-transfer rate, and k_{loss} represents competing loss processes. The electron-transfer rate can be calculated from the electronic coupling, reorganization energy, and driving force:

$$k_{\mathrm{ET}} = \frac{2\pi}{\hbar}|J|^2 \frac{1}{\sqrt{4\pi\lambda k_B T}} \exp\left[-\frac{(\Delta G + \lambda)^2}{4\lambda k_B T}\right]. \tag{2}$$

In a future implementation, QPCS could be used to screen different molecular geometries, donor–acceptor configurations, and candidate photoactive materials, rather than optimizing excitation energy alone. This would extend the current VQE–QSE workflow from electronic-structure prediction toward computational design of efficient photo-induced charge-transfer systems. Such a direction is relevant to SDG 9 through quantum-enabled computational research, and potentially to SDG 7 and SDG 12 through the development and screening of materials for solar-energy conversion and more resource-efficient material discovery.

5 Conclusion

This work established a VQE–QSE workflow for calculating ground- and excited-state properties of ethylene. VQE reproduced the reference active-space ground-state energy, while QSE provided low-lying excitation energies. The results showed a strong dependence of excitation energies on molecular geometry and basis-set choice. With aug-cc-pVDZ, the calculated $\pi \to \pi^*$ excitation was 7.565 eV, corresponding to an excitation-energy deviation of only 0.035 eV from the 7.6 eV reference. In contrast, noise introduced a 177.20 mHa ground-state energy error at the 1× noise level, which was reduced to 94.04 mHa using ZNE-style extrapolation. The donor–acceptor extension further connected the electronic-structure calculations with Marcus theory and Lindblad dynamics for modeling photo-induced charge-transfer processes. As a future direction, the proposed Quantum Photo-Conversion Score could extend this workflow toward the computational screening and design of photoactive donor–acceptor systems for efficient solar-energy conversion.

References

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